

PINAKI CHOUDHURI

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EDUCATION

Adamas University, Kolkata — B.Tech, Computer Science & Engineering (Cyber Security & Forensics)
Expected Graduation: 2027 | Class 10: 86% | Class 12: 70%

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML/CSS, Ruby

Frameworks & Libraries: React JS, Flask, Node.js/Express, Pandas, NumPy, scikit-learn, Playwright

Databases: MongoDB, SQLite, MySQL

DevOps & Cloud: Docker, AWS, Git

Security Tools: Nmap, Nikto, Lynis, John the Ripper

PROJECTS

Hand Gesture Controlled Robotic Arm | *Python, ESP32, Arduino, Bluetooth, IMU* | *Ongoing*

- Engineered a real-time gesture-controlled robotic arm commanding 6 servo motors and 1 stepper motor, achieving positional precision of $\pm 1^\circ$ and $\pm 1.5^\circ$ respectively across all joints.
- Designed a wearable glove interface integrating flex sensors and an MPU6050 IMU to capture 8–15 distinct hand gestures for nuanced robotic manipulation.
- Implemented Bluetooth communication between glove controller and robotic arm with a stable range of ~15m and end-to-end control latency of 80–200ms.
- Coordinated multi-axis servo and stepper motor control to support precise pick-and-place operations in a simulated industrial environment.

Mongo Social — Scalable Social Media Platform | *React, Node.js, Express, MongoDB, REST API*

- Built a full-stack social media platform end-to-end, architecting 10–15 NoSQL collections in MongoDB to handle user profiles, posts, and interaction data.
- Developed a unified RESTful API handling all CRUD operations including user authentication, posting, and social interactions — validated across 3–4 live users.
- Engineered modular analytics pipelines using MongoDB aggregation queries to extract user engagement insights from raw interaction data.

Vroom — Direction-Constrained Ride-Sharing System | *Flask, MongoDB, REST API, Geospatial Mapping*

- Built a full-stack ride-sharing platform using Flask and MongoDB with 6 RESTful API endpoints, incorporating secure authentication and geospatial data management.
- Developed a GPS-based route-matching algorithm using polyline encoding, achieving a 40% same-direction ride pairing rate in prototype testing.
- Integrated external mapping APIs to compute route similarity scores, reducing redundant driver paths across 2–3 concurrent test simulations.

Alvyn v2 — Autonomous Data Analyst Agent | *Python, Flask, LLMs* | [Live Demo](#)

- Engineered and deployed an end-to-end autonomous data analyst agent that handles natural language query → scraping → cleaning → visualization in a single pipeline.
- Exposed a production REST API returning structured JSON insights and base64-encoded plots, designed for sub-second response targets under strict time constraints.
- Architected modular, swappable pipelines for data sourcing, preprocessing, statistical analysis, and dynamic visualization — enabling rapid iteration across data domains.
- Deployed to production on Vercel, demonstrating full-cycle ownership from architecture through live release.

EXPERIENCE

- Competed in 4 hackathons — secured 1st place in a team event and advanced to the final round in all remaining 3 competitions under time-constrained, problem-driven conditions.
- Delivered a technical lecture on cybersecurity and digital forensics to 20+ junior students, covering threat detection methodologies and forensic investigation workflows.
- Contributing to an ongoing research paper on AI-driven prescription recognition (OCR), with 5 months of active research experience in applied machine learning.

NOTABLE

- Earned a 3rd Dan Black Belt in Karate — a decade-long discipline that directly informs how I approach hard engineering problems: structured, persistent, and focused under pressure.
- Operate across the full stack — embedded hardware (ESP32, Arduino), backend APIs (Flask, Node.js), frontend (React), and ML pipelines — without needing to specialize into just one layer.
- Track record of shipping: every project in this resume has a working prototype or live deployment, not just documentation.